Clustering

Assignment Questions

Q1 What is unsupervised learning in the context of machine learning

Ans Unsupervised learning is a type of machine learning where the model is trained on data without labeled responses. Unlike supervised learning, it does not have predefined outputs. The goal is to explore the structure of the data and find hidden patterns or groupings. Common techniques include clustering (like K-means) and dimensionality reduction (like PCA). It is widely used for customer segmentation, anomaly detection, and market basket analysis. The model learns from the input data by identifying similarities and differences. Since there are no correct answers provided, the model discovers the inherent structure on its own. It is useful when labeled data is scarce or unavailable.

Q2 How does K-Means clustering algorithm work?

Ans The K-Means clustering algorithm is an unsupervised learning method used to group data into *K* distinct clusters based on similarity. Here's how it works:

1. Choose the number of clusters (K) you want to form.
2. Initialize K centroids randomly (these are the center points of the clusters).
3. Assign each data point to the nearest centroid based on distance (usually Euclidean distance).
4. Recalculate the centroids by computing the mean of all points assigned to each cluster.
5. Repeat steps 3 and 4 until the centroids no longer change significantly or a maximum number of iterations is reached.
6. The result is K clusters where each point belongs to the cluster with the nearest centroid.

K-Means is fast and efficient but can be sensitive to the choice of K and initial centroids

Q3 Explain the concept of a dendrogram in hierarchical clustering

Ans A dendrogram is a tree-like diagram used to visualize the results of hierarchical clustering. It shows how individual data points are merged into clusters step by step.

- Each leaf of the dendrogram represents a single data point.
- As you move up the diagram, branches merge, indicating the formation of clusters.
- The height at which two branches merge shows the distance or dissimilarity between the clusters being joined.
- By cutting the dendrogram at a certain height, you can decide the number of clusters to form.
- It helps in understanding the structure and relationships within the data.

Dendrograms are especially useful for choosing the optimal number of clusters and analyzing cluster hierarchies.

Q4 What is the main difference between K-Means and Hierarchical Clustering

Ans The main difference between K-Means and Hierarchical Clustering lies in how they form clusters and their approach:

1. K-Means is a partitioning algorithm that requires you to specify the number of clusters (*K*) in advance.
2. Hierarchical Clustering creates a tree-like structure (dendrogram) and does not require a predefined number of clusters. You can choose the number by cutting the dendrogram at a certain level.
3. K-Means is generally faster and more efficient for large datasets.
4. Hierarchical Clustering is more informative but can be computationally expensive for large datasets.
5. K-Means uses centroids to form clusters, while Hierarchical Clustering groups based on distances between data points or clusters.

In summary, K-Means is good for speed and scalability, while Hierarchical Clustering is better for understanding data structure.

Q5 What are the advantages of DBSCAN over K-Means?

Ans DBSCAN (Density-Based Spatial Clustering of Applications with Noise) has several advantages over K-Means, especially in handling complex data:

1. No need to specify the number of clusters in advance, unlike K-Means.

2. Can detect clusters of arbitrary shapes, whereas K-Means tends to form spherical clusters.

3. Robust to outliers and noise, which K-Means often misclassifies or forces into clusters.

4. Works well with unevenly sized and densely packed clusters.

5. Does not rely on centroids, making it better for non-linear cluster boundaries.

Overall, DBSCAN is more flexible and effective for real-world datasets with noise and complex structures.

Q6 When would you use Silhouette Score in clustering ?

Ans You would use the Silhouette Score in clustering to evaluate the quality of the clusters formed by an algorithm. It helps determine how well each data point fits within its assigned cluster compared to other clusters.

Here's when you use it:

1. To assess the performance of clustering algorithms like K-Means, DBSCAN, or Hierarchical Clustering.

2. To choose the optimal number of clusters (K) by comparing silhouette scores for different values of K.

3. To check if the data points are well separated and cohesive within their clusters.

4. To compare different clustering methods on the same dataset.

A higher silhouette score (close to 1) indicates well-formed clusters, while a low or negative score suggests overlapping or incorrect clustering.

Q7 What are the limitations of Hierarchical Clustering ?

Ans Here are the main limitations of Hierarchical Clustering:

1. Scalability issues – It is computationally expensive and not suitable for very large datasets.

2. Irreversible decisions – Once a merge or split is made, it cannot be undone (no backtracking).

3. Sensitive to noise and outliers, which can affect the clustering structure.

4. The choice of linkage method (single, complete, average) can significantly change the results.

5. Difficult to determine the optimal number of clusters without using additional tools like a dendrogram or silhouette score.

6. It may produce imbalanced clusters if the data has varying densities.

Despite these limitations, it is useful for small datasets and understanding hierarchical relationships in data.

Q8 Why is feature scaling important in clustering algorithms like K-Means

Ans Feature scaling is important in clustering algorithms like K-Means because the algorithm relies on distance calculations, usually Euclidean distance, to group data points.

Here’s why scaling matters:

1. Features with larger numerical ranges can dominate the distance calculation, skewing the results.

2. Unscaled data can lead to biased cluster assignments and poor clustering performance.

3. Scaling ensures that all features contribute equally to the clustering process.

4. It helps the algorithm form more accurate and meaningful clusters.

5. Common scaling techniques include Standardization (Z-score) and Min-Max Scaling.

Q9 How does DBSCAN identify noise points ?

Ans DBSCAN (Density-Based Spatial Clustering of Applications with Noise) identifies noise points as data points that do not belong to any cluster based on density criteria.

Here's how it works:

1. DBSCAN uses two parameters: ε (epsilon) – the neighborhood radius, and MinPts – the minimum number of points required to form a dense region.
2. A point is labeled as:
   - Core point: if it has at least MinPts within its ε-neighborhood.
   - Border point: if it's within the ε-neighborhood of a core point but has fewer than MinPts neighbors.
   - Noise point (outlier): if it's not a core or border point.
3. Noise points are isolated and do not meet the density requirements to be part of any cluster.

These noise points are often useful for identifying outliers in the data.

Q10 Define inertia in the context of K-Means

Ans In the context of K-Means clustering, inertia is a metric that measures how tightly the data points in a cluster are grouped around the centroid.

Formally, inertia is defined as the sum of squared distances between each data point and its assigned cluster centroid:

$$\text{Inertia} = \sum_{i=1}^{k} \sum_{x \in C_i} \|x - \mu_i\|^2$$

Where:

- k = number of clusters
- C_i = cluster i
- μ_i = centroid of cluster i
- x = data point in cluster C_i

Key Points:

- Lower inertia means more compact clusters.
- It is used to evaluate and compare clustering performance.
- In the elbow method, inertia is plotted against the number of clusters to help choose the optimal value of K.

Q 11 What is the elbow method in K-Means clustering

Ans The Elbow Method is a technique used to determine the optimal number of clusters (K) in K-Means clustering.

Here's how it works:

1. Run K-Means clustering on the dataset for a range of K values (e.g., from 1 to 10).
2. For each K, calculate the inertia (sum of squared distances between points and their cluster centroids).
3. Plot the inertia values against the number of clusters K.
4. The plot usually shows a sharp decrease in inertia as K increases, then levels off—this point of leveling is called the "elbow."
5. The elbow point indicates a good balance between cluster compactness and complexity, suggesting the optimal number of clusters.

The Elbow Method helps avoid choosing too many or too few clusters by visually identifying where adding more clusters no longer significantly improves the model.

Q12 Describe the concept of "density" in DBSCAN?

In DBSCAN, the concept of density refers to how closely packed data points are in a region of the dataset. It's the key idea used to identify clusters based on point density rather than distance to centroids.

Here's what density means in DBSCAN:

- A region is considered dense if it contains at least a minimum number of points (MinPts) within a specified radius (ε, epsilon).
- Points in dense regions form clusters because they have many neighbors close by.
- Areas with low density—where points don't have enough neighbors—are treated as noise or outliers.
- Core points lie within dense areas, while border points are close to dense regions but don't meet the density criteria themselves.

Q13 Can hierarchical clustering be used on categorical data

Ans Yes, hierarchical clustering can be used on categorical data, but with some considerations:

- Since hierarchical clustering relies on a distance or similarity measure between data points, you need to use a distance metric suitable for categorical data (e.g., Hamming distance, Jaccard similarity, or Matching coefficient), rather than Euclidean distance used for numerical data.
- Some methods convert categorical variables into numerical form (like one-hot encoding) before clustering, but this may increase dimensionality.
- Hierarchical clustering can then proceed using these appropriate similarity measures and linkage methods to group the data.
- It is commonly used for categorical data in areas like customer segmentation or text clustering.

Q14 What does a negative Silhouette Score indicate ?

Ans A negative Silhouette Score indicates that, on average, data points are closer to points in other clusters than to points in their own cluster. This means:

- The clustering assignment is likely incorrect or poor.
- There is overlap between clusters or data points are possibly misclassified.
- The clusters are not well separated and may be mixed or ambiguous.

In practice, a negative silhouette score suggests the need to reconsider the number of clusters, try a different clustering algorithm, or improve data preprocessing.

Q15 Explain the term "linkage criteria" in hierarchical clustering

Ans In hierarchical clustering, the linkage criteria determine how the distance between two clusters is calculated when merging them.

It defines the rule for measuring the similarity or dissimilarity between clusters based on the distances between their points. Common linkage criteria include:

1. Single Linkage: Distance between the closest pair of points (one from each cluster).
2. Complete Linkage: Distance between the farthest pair of points.
3. Average Linkage: Average distance between all pairs of points across the two clusters.
4. Ward's Linkage: Minimizes the total within-cluster variance after merging.

Q16 Why might K-Means clustering perform poorly on data with varying cluster sizes or densities

Ans K-Means clustering may perform poorly on data with varying cluster sizes or densities because:

1. K-Means assumes clusters are spherical and roughly equal in size, so it struggles with clusters that have very different sizes or shapes.
2. It uses distance to centroids to assign points, so clusters with different densities can cause misclassification—points in sparse clusters might be wrongly assigned to denser clusters.
3. Smaller or less dense clusters can be absorbed or ignored by larger, denser clusters.
4. K-Means is sensitive to outliers and noise, which can distort cluster centroids in uneven datasets.

Q17 What are the core parameters in DBSCAN, and how do they influence clustering

Ans The two core parameters in DBSCAN are:

1. ε (Epsilon):
   - Defines the radius of the neighborhood around a point.
   - Determines how close points need to be to each other to be considered neighbors.
   - A small ε may lead to many points labeled as noise, while a large ε can merge distinct clusters.
2. MinPts (Minimum Points):

- The minimum number of points required within the ε-neighborhood for a point to be considered a core point.
- Controls the minimum density needed to form a cluster.
- A higher MinPts makes clusters denser and smaller; a lower MinPts may create larger clusters but more noise.

Together, these parameters control cluster shape, size, and noise detection. Proper tuning of ε and MinPts is crucial for meaningful clustering results.

Q 18 How does K-Means++ improve upon standard K-Means initialization

K-Means++ improves the initialization step of standard K-Means by choosing initial centroids more carefully to speed up convergence and improve clustering quality.

Here’s how it works:

1. **Starts by selecting the first centroid randomly** from the data points.
2. For each subsequent centroid, it selects a new point **with probability proportional to its squared distance from the nearest existing centroid**. This ensures that new centroids are spread out.
3. This method helps avoid poor initializations where centroids are clustered too close together, which can lead to bad clustering results or slow convergence.
4. As a result, K-Means++ often produces **better clusters** and **reduces the number of iterations** needed for the algorithm to converge compared to random initialization in standard K-Means.

Q19 What is agglomerative clustering

Ans Agglomerative clustering is a type of hierarchical clustering that builds clusters from the bottom up. It starts by treating each data point as its own cluster. Then, step by step, it merges the two closest clusters based on a chosen similarity or distance measure. This process continues until all points are merged into a single cluster or until a stopping condition, like a specific number of clusters, is reached. The result is often visualized as a dendrogram, which shows the hierarchy of cluster merges. Agglomerative clustering helps reveal the natural grouping and structure in the data without needing to specify the number of clusters beforehand.

Q20 What makes Silhouette Score a better metric than just inertia for model evaluation?

Ans Silhouette Score is often considered better than inertia for model evaluation in clustering because:

1. Consider both cohesion and separation — it measures how close each point is to its own cluster (cohesion) and how far it is from other clusters (separation).

2. Works well for any clustering algorithm, not just centroid-based ones like K-Means, whereas inertia only measures within-cluster variance.

3. Normalized between -1 and 1, making it easier to interpret: values close to 1 indicate well-clustered data, values near 0 suggest overlapping clusters, and negative values indicate possible misclassification.

4. Helps evaluate cluster quality more comprehensively, while inertia can decrease with more clusters regardless of cluster quality (it always improves with more clusters).

5. Silhouette Score can guide optimal cluster number selection more reliably, as it balances cluster tightness and separation.